

SAT UZBEKISTAN

Telegram @satuzbhunters

1. $(x + 7)$ is a factor of the quadratic function $y = f(x)$, and $x = c$ is a positive zero of the function. If the minimum of the function is $(\frac{63}{8}, k)$, what is the value of c ?
- A) $91/4$ B) $91/2$ C) $91/8$ D) $91/16$
2. For two acute angles, $\angle A$ and $\angle B$, $\cos A = \sin B$. The measures, in degrees, of $\angle A$ and $\angle B$ are $13x + 10$ and $48 + 3x$, respectively. What is the value of x ?
- A) 2 B) 4 C) 5 D) 3
3. A baseball coach charges each athlete \$150 for the first month and \$95 for each additional month. Which of the following functions correctly represents the total revenue $R(m)$, in dollars, for m months and n athletes, where m and n are positive integers?
- A) $R(m) = 95mn + 55$
B) $R(m) = 95mn + 150$
C) $R(m) = 95mn + 55n$
D) $R(m) = 95mn + 150n$
4. In the given system of equations, p is a constant. The graphs of the equations in the given system intersect at exactly one point, (x, y) in the xy -plane. Which of the following could be the value of x ?
- $$\begin{aligned} y &= -3x^2 - 58 \\ y &= px - 46 \end{aligned}$$
- A) -2
B) -4
C) 12
D) 16
5. $f(x) = -7x + b$
- For the given function $f(x)$, which of the following gives the equation for the function $f(x) + 20$, given that the point $(-4, 0)$ lies on the graph of $f(x) - 20$?
- A) $f(x) = -7x - 28$
B) $f(x) = -7x - 8$
C) $f(x) = -7x + 8$
D) $f(x) = -7x + 12$
6. The positive number a is 480% of the number b , and a is 70% of the number c . If c is $p\%$ of b , which of the following is closest to the value of p ?
- A) 267
B) 336
C) 550
D) 685
7. $f(x) = a^x - b$
- The exponential function given above passes through points $(c, 8)$ and $(2c, 350)$. What is a possible value of b ?
- A) 11 B) 12 C) 6 D) 14
- 8.
- | Cylinder | Volume | Surface Area |
|----------|--------------|--------------|
| A | $1,296\pi$ | $k\pi$ |
| B | $162,000\pi$ | $j\pi$ |
- The table gives the volume of two similar cylinders. If the radius of cylinder A is 9 cm. What is the value of $j - k$?
- Volume = $\pi r^2 h$
Surface area = $2\pi r^2 + 2\pi rh$
- A) 10800 B) 9400 C) 11200 D) 11600
9. For the function f , for every increase of $\frac{1}{2}$ in the value of x , the value of $f(x)$ increases by a factor of c , where c is a constant. Which of the following forms of function f displays the value of c as the base or coefficient?
- A) $f(x) = 56(2)^{6x}$
B) $f(x) = 56(8)^{2x}$
C) $f(x) = 56(64)^x$
D) $f(x) = 56(4,096)^{\frac{x}{2}}$

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10. The quadratic function $ax^2 + 200x + c$ has at least one real solution. What is the greatest possible value of ac ?

A) 10000 B) 10001 C) 9999 D) 11000

11. A polygon has exactly 97 sides. If the measure of each of the 97 interior angles of this polygon is $(180p)^\circ$, what is the value of p ?

A) 95/97 B) 97/95 C) 91/97 D) 85/97

12. A certain town has an area of 5.48 square miles. Which of the following is closest to the area, in square yards, of this town? (1 mile = 1,760 yards)

A) 552
B) 964
C) 9,645
D) 16,974,848

13. One of the two equations in a system of linear equations is given. The system has no solution. Which equation could be the second equation in this system?

$$7x = 105y - 133$$

A) $x = 15y$
B) $\frac{1}{7}x = 15y$
C) $x = 15y - 19$
D) $\frac{1}{7}x = 15y - 19$

14. In triangle ABC , the sum of the measures of angle A and angle B is 90 degrees. The value of $\sin(A)$ is $\frac{2\sqrt{7}}{8}$. What is the value of $\cos(B)$?

A) $\frac{2\sqrt{7}}{8}$
B) $\frac{2\sqrt{7}}{6}$
C) $\frac{6\sqrt{2}}{14}$
D) $\frac{14}{23\sqrt{2}}$

15. What is the value of $\frac{k}{25}$ if the given equation only has one solution?

$$25|x - 4| = k$$

A) -5 only
B) 5 only
C) 5 or -5
D) 0

16. In the given system of equations, k is a constant. If the system has no solution, what is the value of k ?

$$\begin{aligned} \frac{3}{8}x + \frac{7}{5}y &= \frac{9}{7} - \frac{7}{5}y \\ \frac{5}{4}x + \frac{9}{4} &= ky + \frac{4}{3} \end{aligned}$$

A) $-28/3$ B) $-26/3$ C) $-19/3$ D) $-14/3$

17. In the xy -plane, the graph of the given equation is a circle. Which point lies on this circle?

$$(x + 7)^2 + (y - 11)^2 = 25$$

A) $(-7, 11)$
B) $(11, -7)$
C) $(\sqrt{11} + 7, \sqrt{14} - 11)$
D) $(\sqrt{11} - 7, \sqrt{14} + 11)$

18. The area of a triangle is equal to x^2 square inches. The base of the triangle is $5 + 2x$ inches, and the height of the triangle is $x - 2$ inches. What is the value of x ?
- A) 2.5
B) 2.7
C) 5
D) 10
19. A circle in the xy -plane has its center at $(3, -7)$ and has a radius of 12. An equation of this circle is $x^2 + y^2 + ax + by + c = 0$, where a , b , and c are constants. What is the value of c ?
- A) -86 B) -94 C) -88 D) -96
20. Sphere A has a radius of $7x$ and sphere B has a radius of $91x$. The volume of sphere B is how many times the volume of sphere A ?
- A) 2197 B) 2226 C) 2464 D) 2187
21. One gallon of resin costs \$79 and will cover 48 square feet of a countertop. The countertops in a given room have a total surface area of k square feet. Which equation represents the cost c , in dollars, of resin needed to cover the countertops twice?
- A) $c = \frac{48k}{79}$
B) $c = \frac{96k}{79}$
C) $c = 79\left(\frac{k}{24}\right)$
D) $c = 79\left(\frac{k}{96}\right)$
22. For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?
- $$\begin{aligned} 10x + 8y &= 15 \\ 25x + 20y &= 37.5 \end{aligned}$$
- A) $\left(r, -\frac{4r}{5} + \frac{15}{8}\right)$
B) $\left(r, \frac{5r}{4} + \frac{4}{5}\right)$
C) $\left(-\frac{4r}{5} + \frac{3}{2}, r\right)$
D) $\left(\frac{2r}{5} + 10, -\frac{2r}{5} + 25\right)$
23. A researcher selected 2 random samples of employees in a large company to estimate the percentage of employees that planned to vote in favor of a new division. Based on the first sample, the researcher estimated that 68% of the employees would vote in favor, with an associated margin of error of 9.7%. Based on the second sample, the researcher estimated that 84% of the employees would vote in favor, with an associated margin of error of 6.7%. Assuming the margins of error were calculated in the same way, which of the following best explains why the second sample obtained a smaller margin of error than the first sample?
- A) The first sample contained fewer employees than the second sample.
B) The first sample contained more employees than the second sample.
C) The first sample contained a lower percentage of residents that planned to vote in favor of a new division.
D) The first sample contained a higher percentage of residents that planned to vote in favor of a new division.

24. The function f is defined by $f(x) = a^x + b$, where a and b are constants. In the xy -plane, the graph of $y = f(x)$ has an x -intercept at $(3, 0)$ and a y -intercept at $(0, -342)$. What is the value of $a + b$?
- A) -336 B) -330 C) -342 D) -348
25. 684 is $p\%$ greater than 9. What is the value of p ?
- A) 7500 B) 7800 C) 8100 D) 8400
26. Two numbers a and b , are each greater than zero, and the fifth root of a is equal to the cube root of b . For what value of x is a^{8x-4} equal to b ?
- A) $23/40$ B) $17/40$ C) $11/40$ D) $27/40$
27. For the given system of equations, what is the value of $7(j - k)$?
- $$(j - k) - 16(m + l) = 689$$
- $$(j - k) + 12(m + l) = 1235$$
- A) 7007 B) 8011 C) 7784 D) 8240
28. A model initially estimates that there are 24,000 bacteria on a petri dish. 14 days later, the model estimates that there are 96,000 bacteria on the petri dish. Assuming exponential growth, the formula $B = a(2)^{xd}$ gives the estimated number of bacteria on a petri dish, where a and x are constants and B is the number of bacteria on the petri dish d days after the initial measurement. What is the value of x ?
- A) $\frac{1}{14}$
 B) $\frac{1}{7}$
 C) 7
 D) 14
29. How many liters of a 30% chlorine solution must be added to 12 liters of a 10% chlorine solution to obtain a 15% chlorine solution?
- A) 4 B) 400 C) 0.4 D) 40
30. In the xy -plane, there are 3 points a , b , and c . Point a has coordinates of $(1, 0)$, point b has coordinates of $(0, 0)$ and point c has coordinates of $(-1, 0)$. Which of the following gives a possible angle measure, in radians of $\angle abc$?
- A) $\frac{456\pi}{6}$
 B) $\frac{459\pi}{6}$
 C) $\frac{462\pi}{6}$
 D) $\frac{468\pi}{6}$
31. Two data Sets A and B have 56 values in total. The average of the two data sets is 193. Data set A contains 32 values with an average of 208. What is the average of data set B ?
- A) 173 B) 175 C) 178 D) 181
32. Lines k and l are perpendicular. If the point $(4, m)$ and $(9, m + 3)$ lies on line k and lines k and l intersect at $(9, m + 3)$, which of the following can lie on line l ?
- A) $(m - 6, -4)$
 B) $(m + 9, 24)$
 C) $(21, m - 17)$
 D) $(24, m + 12)$

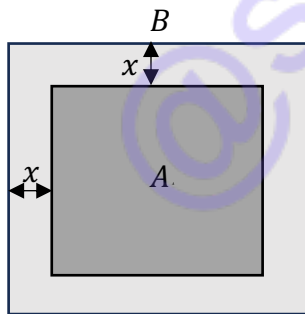
33. Data set A consists of 37 different values that have a minimum of 230, a maximum of 278, a mean of 244, and a standard deviation of 9. The values 230 and 278 are removed from the data set to create data set B . Which of the following statements is true?

- A) Both the mean and standard deviation of data set B are less than those in data set A .
- B) Both the mean and standard deviation of data set B are greater than those in data set A .
- C) The standard deviation of data set B is less than the standard deviation in data set A but the mean of both data sets is the same.
- D) The standard deviation of data set B is greater than the standard deviation in data set A but the mean of both data sets is the same.

34. If $a > 0$, $x^2 + y^2 = a$ and $xy = a - 10$, what is $(x + y)^2$ in terms of a ?

- A) $a - 20$
- B) $2a - 10$
- C) $2a - 20$
- D) $3a - 20$

35. Rectangle A has a width of 50 feet and a length of 45 feet and is located inside rectangle B . A uniform distance x , in feet, separates the sides of rectangle A from rectangle B . The area of rectangle B is 2646 square feet, what is the value of x ?



- A) 2 B) 1 C) 3 D) 4

36. $f(x) = 200(b)^x$. When x increases by 1, $f(x)$ increases by $c\%$. Which of the following expresses c in terms of b ?

- A) $c = \frac{200b}{100}$
- B) $c = \frac{2b}{100}$
- C) $c = (b + 1)100$
- D) $c = (b - 1)100$

37. A 34 pound dog eats two types of canned food: chicken and beef. The recommended amount of the chicken food is 1.25 cans per 16 pounds a dog weighs per day. The recommended amount of canned beef food is 0.85 cans per 23 pounds a dog weighs per day. If c is the number of cans of chicken food and b is the number of cans of beef food a 34 pound dog eats in a given day. Which equation describes all possible values of c and b ?

- A) $\frac{1.25}{16}c + \frac{0.85}{23}b = 34$
- B) $\frac{16}{1.25}c + \frac{23}{0.85}b = 34$
- C) $\frac{1.25}{16}b + \frac{0.85}{23}c = 34$
- D) $\frac{16}{1.25}b + \frac{23}{0.85}c = 34$

38. A quadratic function models the height, in feet, of an object above the ground in terms of time, in seconds, after the object is launched. The model indicates the object launched from the ground (0 feet) and reaches its maximum height of 576 feet above the ground 12 seconds after being launched. Based on the model, what is the height, in feet, of the object above the ground 16 seconds after being launched?

- A) 384
- B) 432
- C) 480
- D) 512

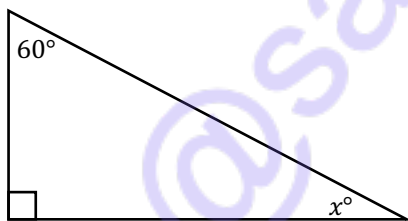
39. A small business owner budgets \$3,600 to purchase timber each year. The owner must purchase a minimum of 345 slats of timber each year to maintain the discounted pricing. If the owner pays \$7.25 to purchase each slat of pine pulpwood and \$24.75 to purchase each slat of pine sawtimber, what is the maximum number of slats of pine sawtimber the owner can purchase to stay within the budget and maintain the discounted pricing?

A) 62 B) 64 C) 66 D) 68

40. A 180 gram metal alloy is 50% aluminum. It contains a metal alloy that is 30% aluminum and a second metal alloy made up of 60% aluminum. What is the mass of the second metal alloy?

A) 120 B) 140 C) 160 D) 180

41. In the given figure, the hypotenuse is 64 what is the $\cos x$?



Note: Figure not drawn to scale.

- A) $\frac{\sqrt{3}}{2}$
 B) $\frac{1}{2}$
 C) $\frac{2\sqrt{3}}{3}$
 D) 2

42. The table shows the distribution of 88 basketball players and their positions on a basketball club. Each player is categorized in one position.

Position	Frequency
Point guard	26
Shooting guard	16
Small forward	11
Power forward or center	35

If one of the 88 players is selected at random, the probability of selecting a player who is categorized as a power forward, given the player is not categorized as a shooting guard, is $\frac{3}{12}$. How many of these players are categorized as a center?

A) 17 B) 21 C) 24 D) 27

43. A quadratic function models the height, in feet, of an object above the ground in terms of time, in seconds, after the object is launched off an elevated surface. The model indicates that at a time of 9 seconds, the object is 294 feet above the ground. At a time of 13 seconds, the object is 310 feet above the ground. If the object was at a height of 24 feet when it was launched, what is the height, in feet, of the object above ground 18 seconds after being launched?

- A) 96
 B) 192
 C) 222
 D) 240

44. A rectangular pyramid has a square base with an area of 324 square meters. What is the surface area, in square meters, of one of the triangular faces if the rectangular pyramid has a volume of 4,320 cubic meters?

A) 369 B) 361 C) 321 D) 324